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SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Code of Conduct 192524358

I AbdulRazakShaik (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: AbdulRazakShaik

C01- AT 1

Course Code : CSA1510

Course : Cloud Computing

Shaik Abdul Razak

(1.) Define Cloud Computing and Explain any four characteristics that make it different from traditional on-premise computing.

A. Cloud Computing:-

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking and software over the internet on demand.

Characteristics:-

- On-demand Self service - users can access resource whenever needed.
- Broad network access - services are available over the internet from different devices.
- Resource Pooling → Resources are shared among multiple users.
- Rapid Elasticity - Resources can be increased or decreased quickly.

(Q.) Trace the Evolution of Cloud Computing from hardware Evolution to internet software Evolution, and show how these changes Enabled modern cloud platforms.

Evolution of cloud computing:-

- Hardware Evolution : Mainframes → Personal Computers → Distributed Systems.
- Networking Evolution : LAN → WAN → Internet.
- Software Evolution : Standalone Software → Client server → Web applications → cloud applications.
- These advancements Enabled Scalable, internet-based cloud platforms.

Explain the role of Server virtualization in cloud Computing, Differentiate between virtualization as an enabling technology and cloud computing as a service model.

A. Role of Server Virtualization:-

- Server Virtualization: Divides one physical server into multiple virtual machines, improving resource utilization.
- Virtualization:- Technology that creates virtual resources.
- cloud Computing: Service model that delivers virtualized resources over the Internet.

(4) Discuss the role of data centers in delivering cloud services.

A. Roles of Data centers:-

- Data Centers are facilities that store cloud servers, storage devices, networking equipment and security systems.
- They provide high availability, scalability, backup and reliable cloud services.

(5) Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

A.

Service Model	Description	Example
IaaS	provides virtual infrastructure	Amazon EC2
PaaS	Provides platform for application development	Google App Engine
SaaS	provides software over the internet	Gmail
XaaS	Any service delivered through the cloud	microsoft 365